

METRO TICKET BOOKING AND SMART METRO CARD RECHARGE SYSTEM USING SERVICENOW

PROJECT DOCUMENTATION

Name: SUGAPRIYA A

1. INTRODUCTION

Metro transportation is one of the most widely used public transportation systems in urban areas. As the number of passengers increases, traditional methods of purchasing tickets and recharging metro cards can result in queues, delays, manual data entry, and difficulty in managing passenger requests. A digital and automated solution can significantly improve the overall passenger experience.

The Metro Ticket Booking and Smart Metro Card Recharge System is developed using ServiceNow to provide a centralized platform for managing metro-related services. The application enables passengers to submit ticket booking and smart card recharge requests through a structured Service Catalog and Service Portal. ServiceNow components such as custom tables, catalog items, variables, UI policies, client scripts, workflows, and access control rules are used to create an integrated application.

The project focuses on creating a simple, secure, and efficient service where passenger information can be collected systematically and requests can be processed through automated workflows. The solution also provides administrators with a structured way to manage station information and passenger transaction data.

2. PROJECT OBJECTIVES

- To provide a digital platform for metro ticket booking and smart metro card recharge.
- To reduce dependency on manual ticket counters and minimize passenger waiting time.
- To maintain metro station information in a centralized ServiceNow database.
- To collect passenger and transaction details through a structured Service Catalog form.
- To dynamically control form fields based on the service selected by the passenger.
- To automate request processing using ServiceNow workflow capabilities.
- To protect application data using roles and Access Control Lists.
- To provide a user-friendly Service Portal experience for metro passengers.

3. SYSTEM OVERVIEW

The system is designed around two primary passenger services: Metro Ticket Booking and Smart Metro Card Recharge. When a passenger accesses the Service Portal, the passenger can select the required service and provide the necessary information. The catalog form dynamically adapts according to the selected option. After submission, ServiceNow creates and manages the request through the configured automation and workflow process.

The application separates master station information from passenger transaction information. This improves data organization and allows station records to be maintained independently while being referenced from the booking form.

4. DATABASE DESIGN AND TABLE CREATION

Two custom tables were created in the Global scope to support the application. The first table is named Metro Station's Details. It functions as the master table for storing information about metro stations. The second table is named Metro Database, which is used to store passenger booking and recharge-related information.

The Metro Station's Details table provides a single source of station information. Instead of allowing users to enter station names manually, the booking form can use reference information from the station table. This improves consistency and prevents spelling variations in station names.

The Metro Database table is designed to capture operational information related to passenger requests. It can store details such as the selected service, source station, destination station, journey type, passenger count, passenger type, fare, smart card information, recharge amount, payment mode, and request-related information.

5. FIELDS AND DATA STRUCTURE

After creating the two tables, appropriate fields were configured according to the requirements of the metro application. The Metro Station's Details table contains station-specific information such as station name, station code, metro line, and station status. These fields help administrators maintain an organized station master.

The Metro Database table contains fields related to the passenger transaction. Important fields include the starting station, destination station, type of journey, number of passengers, passenger type, amount for a single journey, amount including return journey, smart card number, smart card name, recharge amount, and mode of payment.

Reference fields are particularly useful in this project because they allow station information to be selected from existing records rather than entered as free text. This establishes a relationship between the station master table and the transaction data and improves data accuracy.

6. SERVICE CATALOG ITEM

A Service Catalog Item named Book a Metro Ticket was created as the main user-facing service. The Catalog Item provides a structured form through which passengers can request metro services. Instead of navigating through multiple independent forms, passengers can access the required options from a single service interface.

The main service selection variable contains choices such as None, Recharge Metro Card, and Book QR Ticket. The choice values are configured separately from their display order. For example, the display order determines how the choices are presented to the user, while the value is used internally by scripts and policies to identify the selected option.

The Catalog Item acts as the bridge between the passenger-facing Service Portal and the ServiceNow back-end processing. Once the passenger completes the form and submits it, the request becomes part of the ServiceNow request management process.

7. CATALOG VARIABLES

Multiple catalog variables were configured to collect the information required for ticket booking and smart card recharge. The variables include the service selection, Starting From, Going To, Type of Journey, Number of Passengers, Passenger Type, fare-related fields, Smart Card Number, Smart Card Name, Recharge Amount, and Mode of Payment.

Variables are important because they make the catalog item flexible and allow information to be captured in a structured manner. Variables can also be configured with appropriate data types, reference fields, mandatory settings, and default values depending on the requirement.

For station selection, reference-based variables can be used so that passengers select an existing station record. This provides a better user experience and maintains consistency with the Metro Station's Details table.

8. UI POLICIES

Catalog UI Policies were implemented to make the booking form dynamic. The purpose of a UI Policy is to control the visibility, mandatory status, and enabled state of catalog variables based on defined conditions.

For example, when the passenger selects Recharge Metro Card, the smart card number, smart card name, recharge amount, and payment-related fields can be displayed and made mandatory. When the passenger selects the ticket booking option, the relevant journey and passenger fields are displayed instead. Unrelated fields can be hidden.

This approach makes the form cleaner and easier to use because passengers only see the information relevant to the service they have selected. It also reduces the possibility of submitting incomplete or irrelevant information.

9. CATALOG CLIENT SCRIPTS

Catalog Client Scripts were used to add client-side behavior and validation to the catalog form. These scripts can respond to changes in variables, validate values, and perform calculations or dynamic field updates.

For example, the selected service can be checked using its internal choice value, while station and journey information can be retrieved from the form. Client-side logic can also support fare calculations and ensure that the form behaves correctly before submission.

The combination of UI Policies and Client Scripts provides both declarative and scripted control over the catalog form. UI Policies handle straightforward field behavior, while Client Scripts provide additional flexibility for validation and dynamic processing.

10. WORKFLOW AND AUTOMATION

Workflow is an important part of the implementation because the project is not limited to collecting information. After a passenger submits a request, the request must be processed in a systematic manner. The ServiceNow workflow manages the request lifecycle and automates the required processing activities.

The workflow can begin when a booking or recharge request is submitted. It can then move the request through appropriate stages such as request processing, approval where required, task handling, and completion. The workflow provides a consistent sequence of activities and reduces the need for manual intervention.

Automation improves operational efficiency because every request follows a predefined process. Administrators can monitor the status of requests and ensure that pending activities are handled. This makes the system more suitable for real-world service management scenarios.

11. SECURITY AND ACCESS CONTROL

Security was considered as part of the application design. Access Control Lists, commonly known as ACLs, can be configured to restrict access to tables and records based on roles. This prevents unauthorized users from directly accessing sensitive application data.

Role-based access ensures that different users can perform only the activities permitted to them. Passenger-facing users can submit service requests through the portal, while authorized administrators can manage station and transaction information. This separation improves data security and supports controlled application administration.

12. SERVICE PORTAL IMPLEMENTATION

The Service Portal provides the passenger-facing interface for the application. It presents the Metro Ticket Booking service in a simple and accessible format so that users can select their required service and enter the necessary details.

The portal works together with the Catalog Item, variables, UI Policies, Client Scripts, tables, and workflow. When a passenger interacts with the form, the configured logic dynamically responds to the selected service. Once the request is submitted, ServiceNow handles the request using the configured back-end components.

The Service Portal therefore acts as the entry point of the complete solution, while the ServiceNow platform provides the database, business logic, automation, and security behind it.

13. IMPLEMENTATION FLOW

The overall implementation follows a structured sequence. First, the requirements were analyzed and the required metro services were identified. Next, the Metro Station's Details and Metro Database tables were created with their required fields. The Service Catalog Item was then configured with the required variables.

After that, UI Policies and Catalog Client Scripts were implemented to control the form behavior and validation. Workflow automation was configured to manage the request lifecycle. Finally, ACLs and role-based access were considered to secure the application, and the complete solution was made available through the Service Portal.

The resulting architecture connects the passenger interface with the ServiceNow platform in a systematic way: the passenger enters information through the catalog form, the form logic validates and controls the information, the request is stored and processed, the workflow manages its lifecycle, and access controls protect the underlying data.

14. KEY BENEFITS

- Reduced manual effort and passenger waiting time.
- Centralized management of metro station and transaction information.
- Dynamic and user-friendly catalog forms.
- Automated processing of booking and recharge requests.
- Improved consistency through reference-based station selection.
- Controlled access to application data using ACLs and roles.
- Scalable ServiceNow architecture that can support additional metro services.

15. CONCLUSION

The Metro Ticket Booking and Smart Metro Card Recharge System demonstrates how ServiceNow can be used to develop a practical digital transportation service. By combining custom tables, structured fields, Service Catalog, variables, UI Policies, Client Scripts, workflow automation, Service Portal, and security controls, the project provides an integrated approach to managing metro passenger services.

The solution improves the passenger experience while also giving administrators a structured and automated platform for handling service requests. The same architecture can be extended in the future with additional services, notifications, analytics, mobile support, intelligent recommendations, and other smart transportation capabilities.